

Producer statement cover sheet for structural glass balustrade systems

One cover sheet must be provided for each balustrade system

PART 1: TO BE COMPLETED BY THE DESIGN PROFESSIONAL / MANUFACTURER SUPPLYING THE GLASS SYSTEM

Cover sheet issued on behalf of:	Royal Glass
In respect of (system type):	Unex Ascot Style
Supplied to:	Auckland Council
Client / designers name:	
Description of work: <i>E.g. New dwelling</i>	
Address of property:	

PART 2: CONSTRUCTION DETAILS *(tick boxes that apply)*

Deck / balcony / stairs construction:	☐ Timber	☐ Precast	☐ Concrete	☐ Steel	☐ <i>please specify</i>	
Location of deck / balcony / stairs:	☐ Internal	☐ External	☐ <i>please specify</i>			
Deck / balcony / stairs:	☐ New	☐ Existing	Cladding:	☐ Direct fixed	☐ Cavity	☐ N/A

PART 3: GLASS DETAILS *(tick boxes that apply)*

Type:	☐ Toughened	☐ Toughened laminated	☐ Toughened laminated with stiff interlayer
Thickness:	(mm)	Width of pane:	(m)
Height above floor:	(m)	Height above fixing:	(m)
Note: If barrier extends more than 1.2m above fixing point, test reports must be provided			
Supplier:	Royal Glass		

PART 4: BARRIER DETAILS

System name:	Unex Ascot Style
Drawings (ref page number)	Technical Catalogue pages: SS.60.12T, FS.2T.22.60, AS.60.12T
Engineers name:	Dale Donovan
Hardware (type):	Aluminium posts, and glass clamps

Drawings must show the type of hardware being used to support the glass (i.e. channel, discs, etc.)

PART 5: SAFETY FEATURES (tick all boxes that apply)

- | | |
|-------------------------------------|---|
| ☐ | Continuous interlinking rail (show how fixed to building and each pane of glass) |
| ☒ | Continuous interlinking top cap (show how fixed to building and each pane of glass) |
| ☐ | Clamps, brackets or point fixings (show how fixed to building and each pane of glass) |
| ☐ | Framed with structural sealant min. 2 edges (show how fixed to building and each pane of glass) |
| ☐ | Glass with stiff interlayer |
| ☐ | Two or three glass edges supported by continuous clamps |
| ☒ | Structural post to fix interlinking rail to deck |
| ☐ | Other (specify) |

Engineers name:

Dale Donovan

Drawings (ref page number)

Comments:

PART 6: COVER SHEET COMPLETED BY

Name:

Roxy Huang

Signature:



Date:

Contact number:

0800 769 254

Email:

info@royalglass.co.nz

BALUSTRADE PS1 COVER SHEET

Date:

Product | Fixture:

Ascot(VRRTop) | FS.1T.02.00-Dry Timber, Top Fix,
90mm CRS

Indicative Linear Meterage:

Project Name:

Site Address:

Legal Description:

SPECIFIER DETAILS

**Area where balustrade is
installed:**

Company:

Phone:

Email:

Website:

Royal Glass
0800 769 254
support@royalglass.co.nz
www.royalglass.co.nz

NOTE: The attached PS1 covers the relevant balustrade system for multiple sites and applications. The engineer's signature date relates to the calculation of this data. The effective date of this individual PS1 is as shown on this cover sheet.

unex.co.nz

UNEX
ARCHITECTURAL BARRIERS

BALUSTRADES SCREENS POOL FENCES HANDRAILS



association of
consulting and
engineering



ISSUE: 02

PRODUCER STATEMENT – PS1 DESIGN

BUILDING CODE CLAUSE(S): B1/F2/F4

JOB NUMBER: 2130235

ISSUED BY: Kirk Roberts Consulting Engineers Ltd
(Engineering Design Firm)

TO: Unex Systems (NZ) Ltd
(Owner/Developer)

TO BE SUPPLIED TO: To Whom It may Concern
(Building Consent Authority)

Balustrade styles as supplied by Unex System (NZ) Ltd (see list enclosed). Construction details are to be in accordance
IN RESPECT OF: with the UNEX Fabricators' Manuals – Framed Baluster v13.1. The maximum post spacing must not exceed the
distance given in the manual referred to above.

(Description of Building Work)

AT: Various
(Address, Town/City)

LEGAL DESCRIPTION:

N/A ☒

We have been engaged by the owner/developer referred to above to provide (Extent of Engagement):

Refer attached schedule

in respect of the requirements of the Clause(s) of the Building Code specified above for Part only, as specified in the Schedule, of the proposed building work.

The design carried out by us has been prepared in accordance with:

- ☒ Compliance documents issued by the Ministry of Business, Innovation & Employment (Verification method/acceptable solution) Clauses B1/VM1 and B1/AS1. and/or;
- ☐ Alternative solution as per the attached Schedule.

The proposed building work covered by this producer statement is described on the drawings specified in the Schedule, together with the specification, and other documents set out in the Schedule.

On behalf of the Engineering Design Firm, and subject to:

- Site verification of the following design assumptions: Refer to attached Schedule
- All proprietary products meeting their performance specification requirements;

I believe on reasonable grounds that:

- the building, if constructed in accordance with the drawings, specifications, and other documents provided or listed in the Schedule, will comply with the relevant provisions of the Building Code and that;
- the persons who have undertaken the design have the necessary competency to do so.

I recommend the Refer Schedule level of **construction monitoring**.

I, (Name of Engineering Design Professional) Jan Jober-Studnicka, am:

- ☒ CPEng number CMEngNZ, CPEng (1013092), Auckland Council Producer Statement Author Register No. 3312
and hold the following qualifications MSc, CMEngNZ, CPEng

The Engineering Design Firm holds a current policy of Professional Indemnity Insurance no less than \$200,000

The Engineering Design Firm is a member of ACE New Zealand.

SIGNED BY (Name of Engineering Design Professional): Jan Jober-Studnicka

(Signature below):

ON BEHALF OF (Engineering Design Firm): Kirk Roberts Consulting Engineers Ltd

Date: 27/02/2026

Note: This statement has been prepared solely for the Building Consent Authority named above and shall not be relied upon by any other person or entity. Any liability in relation to this statement accrues to the Engineering Design Firm only. As a condition of reliance on this statement, the Building Consent Authority accepts that the total maximum amount of liability of any kind arising from this statement and all other statements provided to the Building Consent Authority in relation to this building work, whether in tort or otherwise, is limited to the sum of \$200,000.

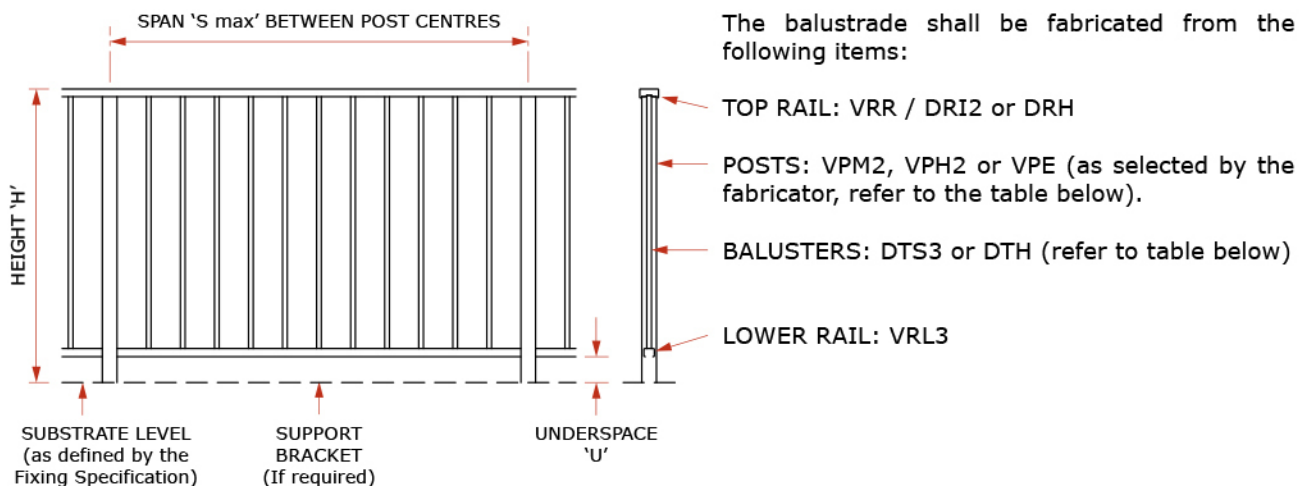
STYLE SPECIFICATIONS

NZBAL-A13.1B | SPEC ID SS.01.12T

'ASCOT' (VRR TOP RAIL)

This specification details the members to be used for this style and the maximum spacing of the posts. A separate specification must be referred to for fixing to the substrate (refer to Chapter 3). Post spacing must not exceed the lesser of the spacing from both Chapter 2 and Chapter 3.

1. Fabrication and Installation are to be in accordance with Assembly Specification AS.01.12T on Page 127 and all relevant portions of this manual.
2. The balustrade shall be constructed such that the geometry of the balustrade shall comply with Acceptable Solutions F4/AS1. This includes the following (but not limited to): The minimum barrier height requirement, the maximum gap allowable and the avoidance of non-complying toeholds.



BALUSTER TYPE AND MAXIMUM POST CENTRES 'S max' (metres) ALWAYS TAKE THE LESSER OF THE VALUE BELOW AND THE VALUE FROM THE FIXING SPECIFICATION								
HEIGHT ⁽³⁾	POST TYPE ⁽²⁾	Line No.	LOADING CLASS ⁽¹⁾					
			N07C/N07R		N03R		Not Preventing Falls	
			BALUSTER TYPE ⁽⁴⁾	SPAN 'S'	BALUSTER TYPE ⁽⁴⁾	SPAN 'S'	BALUSTER TYPE ⁽⁴⁾	SPAN 'S'
1.0	VPM2	1	DTS3	0.99	DTS3	1.71	DTS3	1.91
	VPH2	2	DTS3	1.25	DTS3	1.71	DTS3	1.91
	VPE	3	DTS3	1.50	DTS3	1.87	DTS3	1.91
	RAILS ONLY	4	DTS3	1.91	DTS3	1.91	DTS3	1.91
1.1	VPM2	5	DTS3	0.90	DTS3	1.71	DTS3	1.88
	VPH2	6	DTS3	1.13	DTS3	1.71	DTS3	1.91
	VPE	7	DTS3	1.36	DTS3	1.71	DTS3	1.91
	RAILS ONLY	8	DTS3	1.91	DTS3	1.91	DTS3	1.91
1.2	VPM2	9	DTH	0.74	DTS3	1.59	DTS3	1.55
	VPH2	10	DTH	0.93	DTS3	1.71	DTS3	1.91
	VPE	11	DTH	1.11	DTS3	1.71	DTS3	1.91
	RAILS ONLY	12	DTH	1.91	DTS3	1.91	DTS3	1.91

1. LOADING CLASS: Refer to Page 213 of this MANUAL for the scope of the Loading Class designations.
 2. POST TYPES: Refer to Chapter 1 for details. "RAILS ONLY" gives the maximum span of the rails when fixed in single panels between walls or columns designed to provide adequate support to the rails.
 3. HEIGHT 'H': is the overall height of the balustrade above the substrate level shown. Interpolate for Heights between those shown.
 4. BALUSTER TYPE: Refer to Chapter 1 for details.

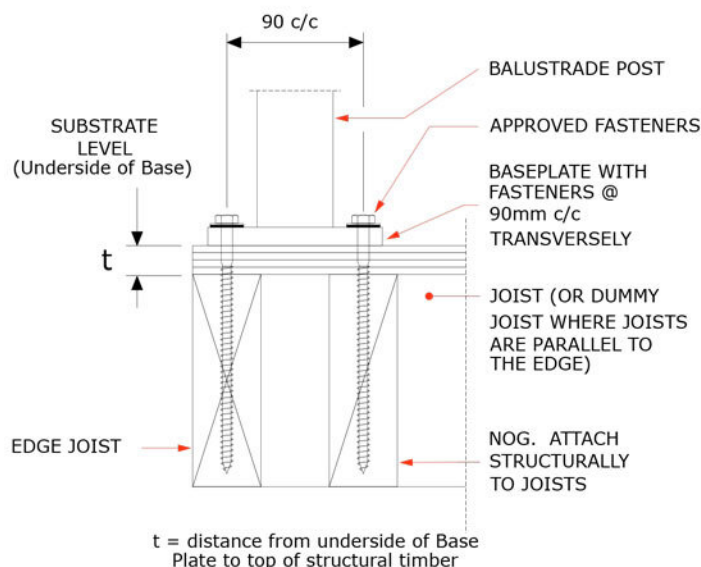
Specifications subject to change without notice

FIXING SPECIFICATIONS

NZBAL-A13.1B | SPEC ID FS.1T.02.00

DRY TIMBER - TOP FIXING, 90MM CRS

Refer to all notes on Pages 83 and 84 which shall apply to this specification and the relevant pages in Chapter 5 Installation Guides. Refer also to Chapter 2 for the Style Specification.



- For details of approved fasteners refer to Note 3 on Page 83.
- Washers to be fitted under screw and bolt heads shall be as follows
 - For 6mm fasteners - washer supplied with fasteners.
 - For FC8-165 fasteners - washer supplied with fasteners.
 - For 8mm bolts - 22mm O.D. S/S washer (Part No. FW8-22) with a polymer washer (Part No. FWP8-22G) between the S/S and the aluminium.
 - For Washers bearing against timber use 50 x 50 x 3mm stainless steel washers Part No FW10-50SQ.
- Substrate design including waterproofing and the structural design of the timber substrate and its connections are not included in this specification and must be carried out by others.

 MAXIMUM POST CENTRES 'S max' (metres) ALWAYS TAKE THE LESSER OF THE VALUE BELOW AND THE VALUE FROM THE STYLE SPECIFICATION																								
Height ⁽³⁾	Baseplate Size D x W	Fasteners - Qty and Type(2)	t' (See dia- gram)	Line No.	LOADING CLASS ⁽¹⁾																			
					N07C/N07R								N03R	Not Preventing Falls										
					Design Wind Speed ⁽⁴⁾									Design Wind Speed ⁽⁴⁾										
					VH	EH								M	H					VH			EH	
					50	52	54	56	58	60	62	64		38	40	42	44	46	48	50	52	54	56	
1.0	115x105	4 x FC8-165	19	1	1.72	1.72	1.72	1.68	1.57	1.46	1.37	1.29	3.68	3.65	3.29	2.99	2.72	2.49	2.29	2.11	1.95	1.81	1.68	
	115x105	4 x FC8-165	25	2	1.63	1.63	1.63	1.58	1.47	1.38	1.29	1.21	3.48	3.43	3.10	2.81	2.56	2.34	2.15	1.98	1.83	1.70	1.58	
	115x105	4 x FC8-165	32	3	1.52	1.52	1.52	1.47	1.37	1.28	1.20	1.13	3.26	3.19	2.88	2.61	2.38	2.18	2.00	1.84	1.70	1.58	1.47	
	115x105	4 x M8 Bolts	N/A	4	2.40	2.40	2.40	2.32	2.16	2.02	1.89	1.77	4.62	4.62	4.54	4.12	3.75	3.43	3.15	2.91	2.69	2.49	2.32	
1.1	115x105	4 x FC8-165	19	5	1.56	1.56	1.50	1.39	1.30	1.21	1.14	1.07	3.35	3.03	2.73	2.48	2.26	2.06	1.90	1.75	1.62	1.50	1.39	
	115x105	4 x FC8-165	25	6	1.48	1.48	1.41	1.31	1.22	1.14	1.07	1.01	3.17	2.85	2.57	2.33	2.13	1.95	1.79	1.65	1.52	1.41	1.31	
	115x105	4 x FC8-165	32	7	1.39	1.39	1.31	1.22	1.14	1.06	1.00	0.94	2.97	2.65	2.39	2.17	1.98	1.81	1.66	1.53	1.42	1.31	1.22	
	115x105	4 x M8 Bolts	N/A	8	2.19	2.19	2.07	1.93	1.79	1.68	1.57	1.47	4.62	4.18	3.77	3.42	3.12	2.85	2.62	2.41	2.23	2.07	1.93	
1.2	115x105	4 x FC8-165	19	9	1.44	1.36	1.26	1.17	1.09	1.02	0.96	0.90	3.08	2.55	2.30	2.09	1.90	1.74	1.60	1.47	1.36	1.26	1.17	
	115x105	4 x FC8-165	25	10	1.36	1.28	1.19	1.11	1.03	0.96	0.90	0.85	2.92	2.40	2.17	1.97	1.79	1.64	1.51	1.39	1.28	1.19	1.11	
	115x105	4 x FC8-165	32	11	1.27	1.20	1.11	1.03	0.96	0.90	0.84	0.79	2.73	2.24	2.02	1.83	1.67	1.53	1.40	1.29	1.20	1.11	1.03	
	115x105	4 x M8 Bolts	N/A	12	2.01	1.88	1.75	1.63	1.52	1.42	1.33	1.24	4.30	3.53	3.19	2.89	2.63	2.41	2.21	2.04	1.88	1.75	1.63	

1. LOADING CLASS: Refer to Page 206 of this Manual for the scope of the Loading Class designations.

2. FASTENER DESIGNATIONS: beginning with 'F' are part numbers for fasteners supplied by UNEX eg. FC8-165: FC = Coach Screw Stainless Steel. 8 = 8mm diameter, 165 = length in mm; Substitution with other fasteners is not permitted.

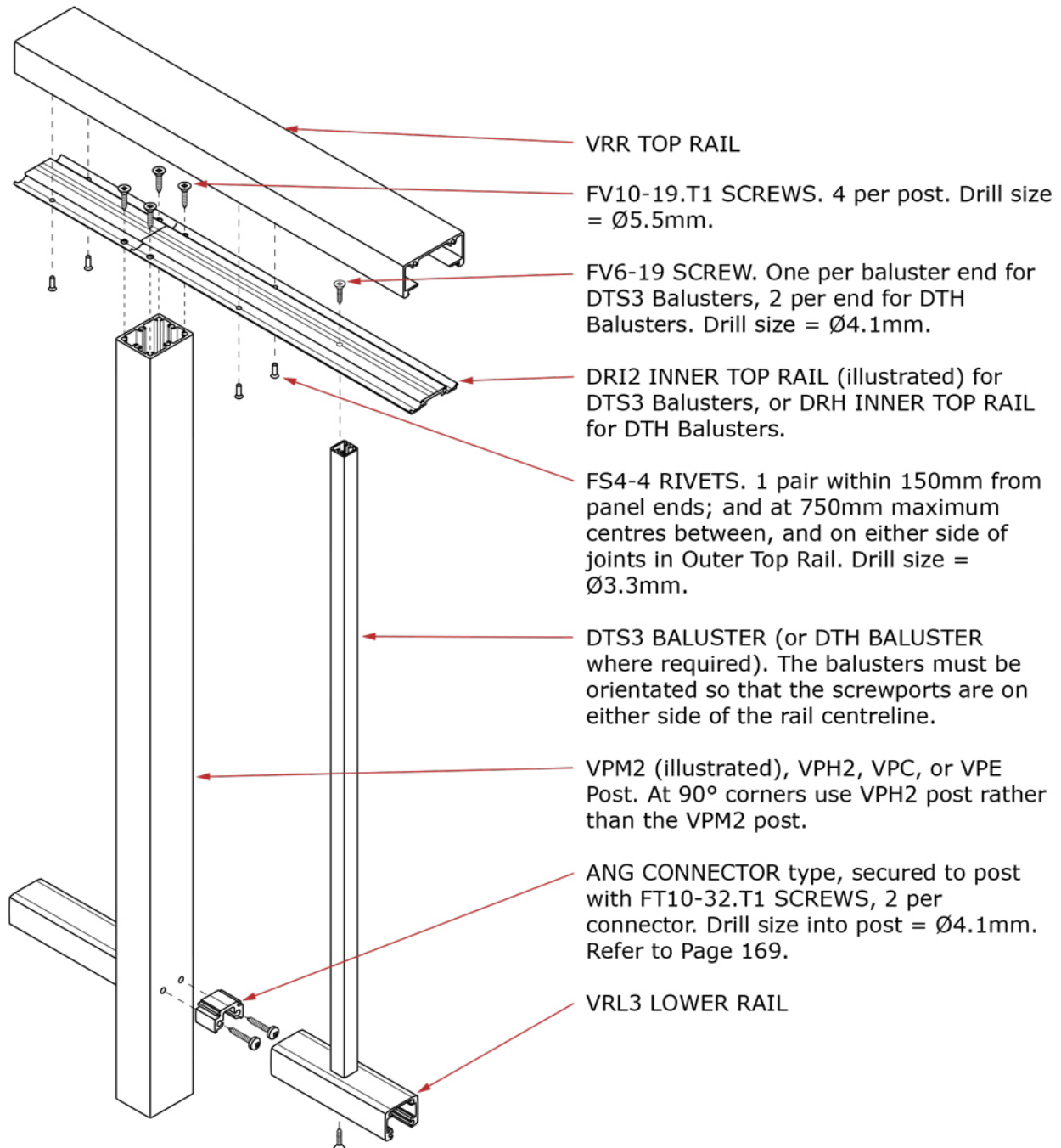
3. HEIGHT 'H': is the overall height of the balustrade above the substrate level shown. Interpolate for Heights between those shown.

4. DESIGN WIND SPEED: in m/s, Refer to Pages 56 to 57 for details of applicable wind codes and the methods for determining the Design Wind Speed.

Specifications subject to change without notice

'ASCOT' (VRR TOP RAIL)

*Refer elsewhere for corners, slopes
and other situations not illustrated here.*



UNEX LOADING CLASSIFICATION

Below is an extract from AS/NZS 1170.1:2002, a standard which is cited in the NZBC. This table gives the minimum imposed actions on barriers for various occupancy types. To the right hand side of this table are the applicable UNEX Loading Classifications (ULC) which are stated in the specifications contained in this catalogue.

AS/NZS 1170.1:2002 TABLE 3.3 MINIMUM IMPOSED ACTIONS FOR BARRIERS							ULC (UNEX Loading Classifi- cations)
Type of occupancy for part of the building or structure	Specific Uses	Top Edge			Infill		
		Horizontal kN/m	Vertical kN/m	Inwards, Outwards, or Downwards kN	Horizontal kPa	Any Direction kN	
A Domestic and residential activities	All areas within or serving exclusively one dwelling including stairs, landings, etc., but excluding external balconies and edges of roofs (see C3)	0.35	0.35	0.6	0.5	0.25	N03R
	Other residential (see C3)	0.75	0.75	0.6	1.0	0.5	N07R
B, E Offices and work areas not included elsewhere including storage areas	Light access stairs and gangways not more than 600mm wide	0.22	0.22	0.6	N/A	N/A	N02
	Fixed platforms, walkways, stairways and ladders for access	0.35	0.35	0.6	N/A	N/A	N03C
	Areas not susceptible to overcrowding in office and institutional buildings also industrial and storage buildings	0.75	0.75	0.6	1.0	0.5	N07C
AREAS WHERE PEOPLE MAY CONGREGATE							
C1/C2 Areas with tables or fixed seating	Areas with fixed seating adjacent to a balustrade, restaurants, bars, etc.	1.5	0.75	0.6	1.5	1.5	N15
C3 Areas without obstacles for moving people not susceptible to over-crowding	Stairs, landings, external balconies, edges of roofs, etc.	0.75	0.75	0.6	1.0	0.5	N07C
C5 Areas susceptible to over-crowding	Theatres, cinemas, grandstands, discheques, bars, clubs, auditoria, shopping malls (see also D), assembly areas, studios, etc	3.0*	0.75	0.6	1.5	1.5	N30 N20 ⁽¹⁾
D Retail areas	All retails areas including public areas of banks/building societies (see C5 for areas where over-crowding may occur)	1.5	0.75	0.6	1.5	1.5	N15

⁽¹⁾ N20 is only applicable where 2.0kN/m horizontal top edge load is acceptable by the Building Consent Authorities.

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